

Answer Key

Recall Edition — check only after you have written every cell.

01 • Capital budgeting decision rules

- (1) NPV measures value added in currency and is always correct for mutually exclusive projects; the others are shortcuts that fail in a specific, predictable way.
One-sentence model • ●●
- (2) Under hard capital rationing, ranking by NPV alone can waste budget
NPV • When it breaks • ●●●
- (3) Ignores every cash flow after the cutoff
Payback • When it breaks • ●●●
- (4) Same as NPV
PI • Key assumption • ●●
- (5) Capital rationing — ranking by value per unit of scarce capital
PI • When to use it • ●●
- (6) Treating a short payback as evidence of profitability
Payback • Common mistake • ●●●
- (7) Discounting at the IRR instead of the WACC
NPV • Common mistake • ●●●
- (8) Reinvestment at the cost of capital
NPV • Key assumption • ●●
- (9) Can rank against NPV when project scales differ
PI • When it breaks • ●●●
- (10) Non-conventional cash flows → multiple or no IRR
IRR • When it breaks • ●●●
- (11) Forgetting that $PI > 1$ and $NPV > 0$ are the same condition
PI • Common mistake • ●●●
- (12) Reinvestment at the IRR itself — usually unrealistic
IRR • Key assumption • ●●
- (13) What the method assumes you do with the money that comes back. NPV assumes you earn the cost of capital on it; IRR assumes you earn the IRR. Every ranking conflict between them traces to that one assumption.
Decisive distinction • ●●●
- (14) If the projects are mutually exclusive or differ in scale or timing, follow NPV.
NPV / IRR • ●●●
- (15) PI is a ratio. Ratios rank by efficiency, not by size — that is the whole difference.
PI / NPV • ●●●
- (16) False for mutually exclusive projects — select on NPV
The project with the higher IRR should be selected. • ●●●
- (17) Not defined when cash flows are non-conventional and no real IRR exists
A project with a positive NPV always has an IRR above the cost of capital. • ●●●
- (18) It corrects for time value but still ignores all cash flows after the cutoff
Discounted payback fixes the flaws of payback. • ●●●

02 • Depreciation methods

- (19) Accelerated methods move expense into early years, so early net income and early carrying amount are lower, and both reverse before the asset's life ends.
One-sentence model • ●●
- (20) Using remaining life for the numerator of the wrong year
Sum-of-years digits • Common mistake • ●●●
- (21) Using it as the default when the vignette states usage-based consumption
Straight line • Common mistake • ●●●
- (22) Subtracted up front
Straight line • Salvage treatment • ●●●
- (23) Subtracting salvage before applying the rate
Double declining balance • Common mistake • ●●●
- (24) Subtracted up front
Sum-of-years digits • Salvage treatment • ●●●
- (25) Depends on usage, not on time
Units of production • Early-year expense • ●●●
- (26) Failing to revise the rate when total estimated units change
Units of production • Common mistake • ●●●
- (27) Whether salvage value enters the calculation at the start. Every method subtracts it up front except DDB, which uses it only as a floor.
Decisive distinction • ●●●
- (28) If salvage is missing from the first-year computation, it is DDB.
DDB / SYD • ●●●
- (29) It only shifts income between periods — the lifetime total is unchanged
Accelerated depreciation reduces total reported income. • ●●●
- (30) It is a change in estimate, applied prospectively
A change in useful life requires restating prior periods. • ●●●